

Population models of *Minecraft* sheep

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1 Introduction

Minecraft is an open-world sandbox video game¹. The game takes place in a procedurally generated world made of cubic voxels, or *blocks*. Among other activities, players can engage in agriculture, including the raising of sheep.

Players can gather blocks of wool by raising a flock of sheep in a pasture (Figure 1). When a sheep is sheared, they must eat grass to regrow their wool. When a grass block is eaten, it converts into dirt, and can regrow if grass spreads from an adjacent grass block.

One may be tempted to pack the pasture as densely as possible to maximize space efficiency. However, if a pasture is packed too densely with sheep, they will deplete the grass, and wool production will be low. On the other hand, if the pasture is not densely packed, wool production will still be low because there are not many sheep. To understand where the happy medium lies, population models were developed for the grass and sheep populations.

1.1 Deterministic vs. Stochastic

Models of biological populations can be broadly divided into *deterministic* and *stochastic*. Deterministic models are simpler to analyze, but stochastic models are generally more informative since biological populations, both in real life and in *Minecraft*, are stochastic processes. In the real world, populations are influenced by random ecological events, and in *Minecraft*, the growth of grass and the appetite of sheep are governed by random number generation.

¹This investigation was conducted in *Minecraft* Java edition version 1.19.4.

Figure 1: A pasture in a meadow, with woolly and sheared sheep, and grass and dirt blocks.



2 Table of symbols

N	Sheep population
A	Pasture size
N_g	Grass population
λ	Grass growth rate
μ	Grass consumption rate
k	Grass growth rate constant
R	Grass consumption rate constant
P	Pasture carrying capacity
p_x	Probability distribution of N_g
q_{xy}	Probability distribution of N_g and N_w

3 Deterministic model

3.1 Grass population

The *Minecraft* physics engine operates in discrete time increments of 0.05 seconds, called *ticks*. Here, this will be approximated as a continuous-time process.

Sheep randomly attempt to eat the block they're standing on. If the sheep and grass in the pasture are uniformly distributed, the grass consumption rate μ is given by:

$$\mu = \frac{NRN_g}{A} \quad (1)$$

Where A is the size of the pasture, N_g is the number of grass blocks in the pasture, N is the number of sheep, and R is the frequency with which an individual sheep attempts to eat.

Grass blocks randomly spread to adjacent dirt blocks. If N_g is small, then grass growth will be limited by the lack of places the grass can spread *from*. If N_g is large, grass growth will be limited by a lack of places for the grass to spread *to*. A simple approach is to assume the growth rate λ has a parabolic dependence on N_g :

$$\lambda = kN_g - jN_g^2$$

The growth rate must be zero for $N_g = 0$ and $N_g = A$, which requires $j = \frac{k}{A}$:

$$\lambda = kN_g - \frac{k}{A}N_g^2 \quad (2)$$

This leaves k as a constant related to the rate at which grass spreads. The overall rate of change of N_g is the difference of λ and μ :

$$\dot{N}_g = \lambda - \mu = \left(k - \frac{NR}{A}\right)N_g - \frac{k}{A}N_g^2 \quad (3)$$

This model, in which $\dot{N}_g$ has quadratic dependence on N_g , is known as the *logistic* model. This model is one of the oldest models for population growth, originally proposed in 1838 [4]. It has since been joined by many more general models [5] which are more useful for real-world ecology, but since we are modelling a video game, we will employ it for its simplicity.

The exact solution [5] of equation 3 is:

$$N_g(t) = \frac{P}{1 + \left(\frac{P}{N_g(0)} - 1\right)e^{-(k - \frac{NR}{A})t}} \quad (4)$$

Where

$$P = A - \frac{NR}{k}$$

Note that the population asymptotically approaches P as $t \rightarrow \infty$. In ecology, P is known as the carrying capacity, the maximum population that a habitat can sustain in equilibrium. This is not to be confused with A , the number of blocks in the pasture and therefore the maximum number of grass blocks it can contain.

Note also that for $N \geq \frac{Ak}{R}$, the carrying capacity is 0.

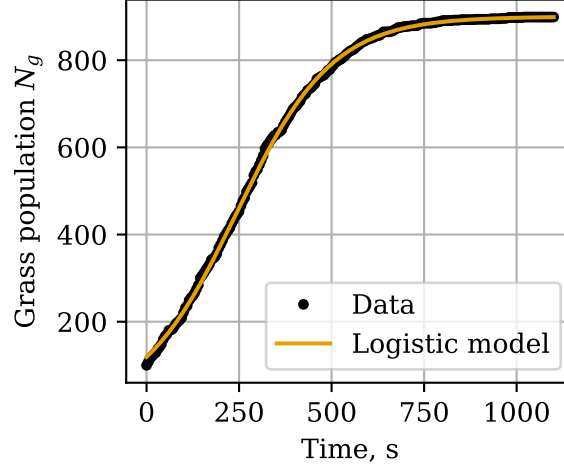
3.2 Experimental determination of rate constants

The actual values of k and R were determined with in-game experiments. R was measured by counting the number of grass blocks eaten by sheep. In 1 hour, 10 sheep ate 320 blocks of grass, which gives $R = 0.00888 \text{ s}^{-1}$.

k was measured by uniformly dispersing 100 grass blocks in a 900-block dirt patch and tracking the grass population over time. The grass growth was then fit to equation 4. This fit gave $k = 0.00772 \text{ s}^{-1}$. More details about these experiments can be found on the author's Github².

²github.com/

Figure 2: Logistic model fit to grass growth data.



3.3 Woolly sheep population

A naïve assumption is that the farmer has superhuman speed, and shears each sheep the instant it grows its wool. In this case, the rate at which wool is gathered³ is equal to μ . Denoting the rate of wool gathering with F , and substituting $N_g = P$ into equation 1 (assuming the pasture has equilibrated):

$$F_{fast} = \mu = NR - \frac{N^2 R^2}{Ak} \quad (5)$$

This is a simple parabolic relationship with a maximum value of $F_{fast} = \frac{Ak}{4}$ at $N = \frac{Ak}{2R}$.

Minecraft players typically don't want to put all of their time into shearing their sheep as quickly as they can. A more realistic assumption is that the farmer shear the sheep periodically and at regular intervals. If the farmer shears the sheep with period T , the rate at which they gather wool is:

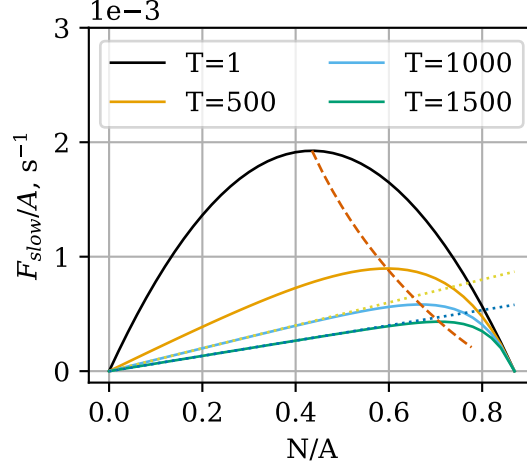
$$F_{slow} = \frac{N_w(T)}{T}$$

Analyzing this model requires analyzing the evolution of N_w with time. A sheep will only regrow wool upon eating if it doesn't already have any, so we should expect that the rate at which sheep are regrowing wool is the rate at which they are eating, μ , times the fraction of sheep that are wool-less:

$$\dot{N}_w = \frac{N - N_w}{N} \mu = (N - N_w) \frac{RN_g}{A} \quad (6)$$

³Shearing a sheep actually yields between 1-3 blocks of wool, but this is out of the player's control, so it is assumed here that a sheep always yields one wool.

Figure 3: Contours of F_{slow} as a function of N at constant T . The dashed line traces the optimal N and F_{slow} as T increases.



Solving equation 6 with the condition $N_w(0) = 0$, and assuming $N_g = P$ (and is therefore time-independent) yields:

$$N_w(T) = N - Ne^{-TR + \frac{TR^2 N}{Ak}}$$

Therefore:

$$F_{slow} = \frac{N - Ne^{-TR + \frac{TR^2 N}{Ak}}}{T} \quad (7)$$

The optimal N can be found by differentiating equation 7 w.r.t. N , setting $\frac{dF_{slow}}{dN} = 0$, and solving for N :

$$N = \frac{Ak}{TR^2} (W(e^{TR+1}) - 1) \quad (8)$$

Where W is the Lambert W function, which satisfies $W(xe^x) = x$. Plugging this back into equation 7, we obtain the value of F_{slow} at optimal N :

$$F_{slow} = \frac{Ak(W(e^{TR+1}) - 1)(1 - e^{-TR + W(e^{TR+1}) - 1})}{R^2 T^2} \quad (9)$$

Figure 3 shows a plot of equations 7-9. As T increases, the optimal sheep population also increases, and the optimal rate of wool gathering decreases. Clearly there is a tradeoff between the frequency with which the farmer shears the flock and the rate at which they gather wool.

For small T , $F_{slow} \approx F_{fast}$. This can be shown quantitatively by employing a first-order approximation for e^x , for $x \approx 0$:

$$e^x \approx x + 1$$

In equation 7, the exponent is small for small values of T , therefore:

$$F_{slow} \approx \frac{N - N(-TR + \frac{TR^2N}{Ak} + 1)}{T} = NR - \frac{N^2R^2}{Ak} = F_{fast}$$

It appears in Figure 3 that for large T and small N , F_{slow} is approximately linear w.r.t. N . Under these conditions, all of the sheep have regrown their wool by the time the farmer comes back, and $N_w(T) \approx N$. The exponent $-TR + \frac{TR^2N}{Ak}$ in equation 7 is large and negative, so $F_{slow} \approx \frac{N-0}{T} = \frac{N}{T}$. Lines of $\frac{N}{T}$ are plotted over top of the curves in Figure 3.

4 Stochastic model

4.1 Grass population

A deterministic model considers population as a single number. A stochastic model considers the *probability distribution* of the population. Here, a stochastic model will be developed for the grass population of the pasture, modelling the grass population as a *Markov process*. The method used here is described in [1], and applied here for the case of the logistic model.

Consider a pasture at time $t = t_0$, with $N_g(t_0) = x$. A short time ago, at time $t = t_0 - \Delta t$, the pasture had $N_g(t_0 - \Delta t)$. The pasture may have arrived at $N_g(t_0) = x$ in one of four ways:

1. $N_g(t_0 - \Delta t) = x$, and neither eating nor growth take place.
2. $N_g(t_0 - \Delta t) = x - 1$, and one grass block regrows.
3. $N_g(t_0 - \Delta t) = x + 1$, and one sheep eats a grass block.
4. Two or more growth and eating events occur.

The probability of each pathway is the probability that the pasture was in the required state $N_g(t_0 - \Delta t)$ times the probability of the event (eating or growth) required to bring the pasture to $N_g(t_0) = x$.

λ and μ are the expected number of grass blocks that regrow or are consumed in unit time, respectively. For a sufficiently short time interval Δt , $\lambda\Delta t$ and $\mu\Delta t$ are the probability that one grass block regrows or is consumed in time Δt , respectively. This gives the probability of nothing happening in time Δt as $(1 - \lambda\Delta t - \mu\Delta t)$. The probability of two or more events is some product of $\lambda\Delta t$ and $\mu\Delta t$, and is therefore dependent on Δt with order 2 or higher.

Let $p_x(t)$ be the probability that the pasture has $N_g = x$ at time t . Then the probability of each path is:

$$\begin{aligned} \text{Pr}(1) &= p_x(t_0 - \Delta t)(1 - \lambda\Delta t - \mu\Delta t) \\ \text{Pr}(2) &= p_{x-1}(t_0 - \Delta t)\lambda\Delta t \\ \text{Pr}(3) &= p_{x+1}(t_0 - \Delta t)\mu\Delta t \\ \text{Pr}(4) &\approx O(\Delta t^2) \end{aligned}$$

The probability of the pasture arriving at $N_g(t_0) = x$ is the sum of the probability of each pathway:

$$\begin{aligned} p_x(t_0) &= p_x(t_0 - \Delta t)(1 - \lambda\Delta t - \mu\Delta t) \\ &\quad + p_{x-1}(t_0 - \Delta t)\lambda\Delta t \\ &\quad + p_{x+1}(t_0 - \Delta t)\mu\Delta t \\ &\quad + O(\Delta t^2) \end{aligned}$$

subtracting $p_x(t_0 - \Delta t)$, dividing by Δt , and taking $\lim_{\Delta t \rightarrow 0}$ converts this into a differential equation:

$$\dot{p}_x(t) = -p_x(t)(\lambda(x) + \mu(x)) + p_{x-1}(t)\lambda(x-1) + p_{x+1}(t)\mu(x+1) \quad (10)$$

Note that $\mu(x)$ simply represents the value of μ with $N_g = x$, and likewise for λ . Note also that the $O(\Delta t^2)$ term has vanished upon taking the limit $\Delta t \rightarrow 0$.

A pasture of size A will have a system of $A + 1$ equations. Systems of equations such as these, which describe the evolution of a probability distribution with time, are known as *Kolmogorov equations* [1]. In general, Kolmogorov equations can be used to describe a *Markov process*, i.e. a process by which a system transitions between multiple possible states. In this case the states are the possible values of N_g .

In vector form, this system is expressed as $\dot{\mathbf{p}}(t) = \mathbf{p}(t)\mathbf{R}$, where $\mathbf{p}(t) = [p_0(t), p_1(t), p_2(t) \dots p_A(t)]$, and $\mathbf{R}$ is a matrix of coefficients, such that the coefficient $r_{i,j}$ relates p_i with $\dot{p}_j$:

$$\begin{aligned} \dot{p}_j &= \sum_i r_{ij} p_i \\ r_{i,j} &= \begin{cases} \lambda(j-1) & \text{for } i = j-1 \\ \mu(j+1) & \text{for } i = j+1 \\ -(\lambda(j) + \mu(j)) & \text{for } i = j \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

This system was solved numerically for pastures of sizes $A = 100, 200$, and 500 , and a range of N values. *Minecraft* pastures typically begin fully covered with grass, so the initial condition will be $\mathbf{p}(0) = [0, 0, 0 \dots 1]$.

Figure 4 compares the expected value of N_g after 10000 seconds with the deterministic carrying capacity. They line up very closely for small sheep densities, but for large sheep densities, the expected N_g is significantly less than P , especially for $A = 100$.

Figure 5 shows the evolution of the distribution of N_g for $N = 40$ and $N = 70$ in a 100-block pasture. The pasture with $N = 40$ approaches a stable distribution, evidenced by the indistinguishable distributions at $t = 1600$ s and $t = 9500$ s. In contrast, the pasture with $N = 70$ continues evolving toward smaller and smaller grass populations, with p_0 increasing dramatically as time proceeds.

Figure 4: Expected value of N_g at $t = 10000$, for $A = 100$ and $A = 500$. P from the deterministic model is shown as a dotted line.

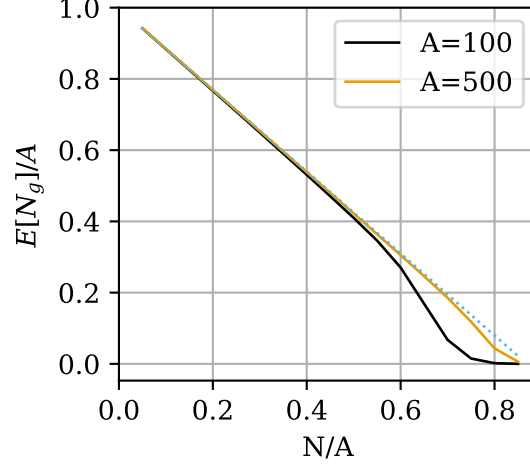


Figure 5: Evolution of N_g distribution for a pasture of size $A = 100$, for (a) $N = 40$ and (b) $N = 70$.

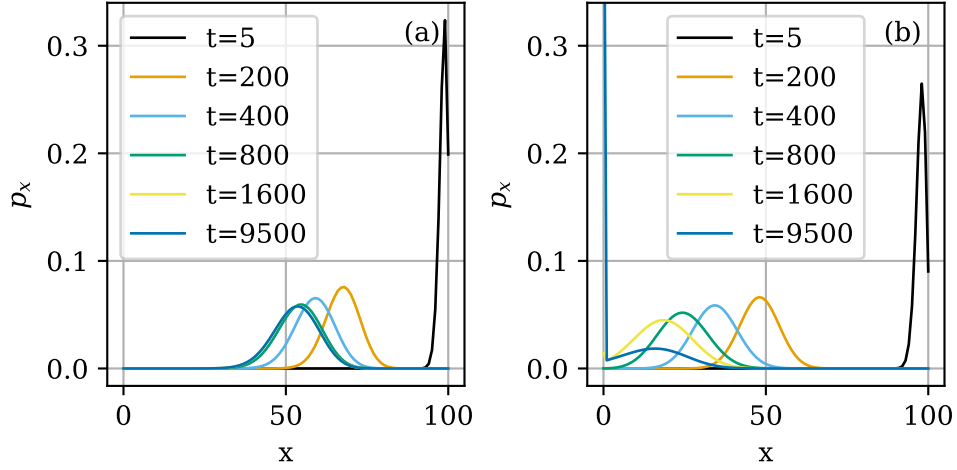
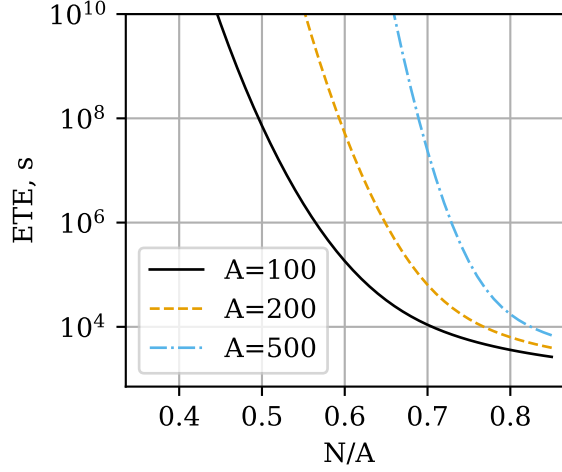


Figure 6: Expected time to extinction for pastures of $A = 100, 200$, and 500 as a function of sheep population density.



The reason for this is evident from the equation for $\dot{p}_0$. Beginning with equation 10 and considering that $p_{-1} = 0$ and $\lambda(0) = \mu(0) = 0$:

$$\dot{p}_0(t) = p_1(t)\mu(1)$$

Since p_1 and $\mu(1)$ are always positive, p_0 is monotonically increasing, bounded above by $p_0 = 1$. Therefore the true equilibrium state is $\mathbf{p} = [1, 0, 0, 0, \dots]$ i.e. extinction. However, even if this is the equilibrium state, the pasture may remain at a different *metastable* (or *quasi-equilibrium*) distribution if the probability of extinction remains negligible across reasonable time scales, as seen in Figure 5a.

Rapid grass extinction at high sheep densities poses a problem, since the deterministic model predicts that the optimal N increases as T increases (Figure 3). Fortunately, an exact solution can be obtained for the expected time to extinction ETE. A mean-residence time matrix can be constructed from the negative of the inverse of the $\mathbf{R}$ matrix for transient states [2] (see also [1] section 6.8.1.2.). Summing across rows of this mean residence time matrix gives the total time the pasture spends with nonzero population.

Figure 6 shows the ETE as a function of N . For all three pasture sizes, populations close to $\frac{Ak}{R} = 0.869$ have ETE on the order of a few hours. As the sheep population density decreases, the ETE increases dramatically - For a 100 block pasture with a sheep density of 0.5, the ETE is over 2 years. ETE also increases dramatically with larger pastures - A 100-block pasture with sheep density 0.7 has ETE of 3 hours, while a 500-block pasture with the same density has ETE of 9 months.

4.2 Woolly sheep population

A model for both the sheep and grass mechanics will involve a bivariate probability distribution, q_{xy} , where:

$$q_{xy}(t) = \Pr(N_g(t) = x, N_w(t) = y)$$

For a pasture to arrive at $N_g(t+0) = x$ and $N_w(t_0) = y$, The possible events are:

- $N_g(t - t_0) = x$, $N_w(t - t_0) = y$, and neither eating nor growth take place.
- $N_g(t - t_0) = x - 1$, $N_w(t - t_0) = y$, and one grass block regrows.
- $N_g(t - t_0) = x + 1$, $N_w(t - t_0) = y$ and one woolly sheep eats grass.
- $N_g(t - t_0) = x + 1$, $N_w(t - t_0) = y - 1$, and one woolless sheep eats grass.
- Two or more growth and eating events occur.

For the two cases involving one block of grass being eaten, the probability will include a term pertaining to the number of sheep who have their wool. By analogy with equation 6, the rate at which woolless sheep eat grass is $\frac{N - N_w}{N} \mu$, and the rate at which woolly sheep eat grass is $\frac{N_w}{N} \mu$. Following the same procedure as with equation 10:

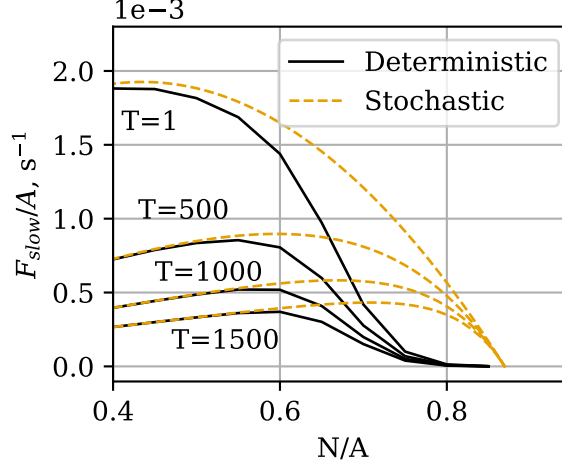
$$\begin{aligned} \dot{q}_{x,y}(t) = & q_{x,y}(t)(-\mu(x) - \lambda(x)) \\ & + q_{x-1,y}(t)\lambda(x-1) \\ & + q_{x+1,y}(t)\mu(x+1)\frac{y}{N} \\ & + q_{x+1,y-1}(t)\mu(x+1)\frac{N - (y-1)}{N} \end{aligned} \quad (11)$$

The resulting system of $(A+1)(N+1)$ equations can be expressed as $\dot{\mathbf{q}}(t) = \mathbf{q}(t)\mathbf{R}$, where $\mathbf{q}(t)$ is a matrix of elements $q_{ij} = q_{xy}(t)$, and $\mathbf{R}$ is a rank-4 tensor:

$$r_{ijkl} = \begin{cases} -\mu(i) - \lambda(i) & \text{for } (i, j) = (k, l) \\ \lambda(k-1) & \text{for } (i, j) = (k-1, l) \\ \mu(k+1)\frac{l}{N} & \text{for } (i, j) = (k+1, l) \\ \mu(k+1)\frac{N-(l-1)}{N} & \text{for } (i, j) = (k+1, l-1) \\ 0 & \text{otherwise} \end{cases}$$

As before, this system can be solved numerically. The initial values of $q_{x,0}$ were set using the values of $p_x(t = 10000)$ from the grass population model. 7 shows F_{slow} as calculated from the results of this model. Similarly to Figure 4, the stochastic model predicts significantly lower F_{slow} than the deterministic model.

Figure 7: F_{slow} calculated from the stochastic model for sheep and grass, for a pasture with $A = 100$. The initial state was obtained from the numerical solution of p_x at $t = 10000$.



The root cause of this deviation is the same as for the grass growth model. The equations for $\dot{q}_{0,y}$ are:

$$\dot{q}_{0,y}(t) = q_{1,y}(t)\mu(1)\frac{y}{N} + q_{1,y-1}(t)\mu(1)\frac{N-(y-1)}{N}$$

Since all terms here are positive, $q_{0,y}$ are monotonically increasing, and given enough time, the pasture will tend to evolve into one of these states and never leave.

The biggest drawback of this model is its computational intensiveness. A model for a pasture with $A = 500$ and $N = 300$ would have $(500+1)(300+1) = 150801$ simultaneous equations, and an $\mathbf{R}$ tensor with $150801^2 = 22740941601$ elements, which occupy 91 GB of memory if stored as 64-bit floats, making computation infeasible for all but top-shelf consumer hardware.

5 Conclusions

The logistic model, and the Kolmogorov equations, have proven useful for predicting the behavior of a pasture in *Minecraft*. However, there are still a few unanswered questions.

First, a pasture in *Minecraft* is often constructed by fencing off an area of grass from a larger meadow (see Figure 1). Grass can spread under fences, effectively introducing an immigration term to the logistic model that is nonzero even when the grass in the pasture is extinct. Such a model is studied in [1] chapter 7, but the deterministic model becomes rather unwieldy, so the effect of immigration was ignored here. However, introducing immigration would have

an important effect on the behavior of the pasture for large N , since $N_g = 0$ is no longer an absorbing state.

Second, time in *Minecraft* is discretized into increments of 0.05 seconds, while we have been modelling the pasture with continuous-time processes. How would a discrete-time model differ from a continuous-time model?

References

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